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Tailoring lithium dicyanoimidazolidine: structure-property relationships

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Abstract

Taking lithium 2-trifluoromethyl-4,5-dicyanoimidazol-1-ide as a parent lithium salt for Li-ion batteries, a systematic property tuning in two series based on variously 2- and 4-substituted 4,5-dicyanoimidazolidine and 2-phenyl-4,5-dicyanoimidazolidine scaffolds is demonstrated. A straightforward synthetic approach afforded fourteen desired derivatives with a systematically evaluated structure, whose properties were further investigated from various perspectives. Stabilization of the imidazolidine anions via the substituent effects has been examined using dissociation constants, Hammett equation, ¹³C NMR shifts and electronic absorption spectra. Solubility in dimethyl carbonate and 1,2-dimethoxyethane has further identified lithium salts with a potential for a practical application and their solutions were further investigated for aggregation phenomena. Compared to the current state-of-the-art and using absorption spectroscopy, a significantly more sensitive and straightforward methodology is presented, which allowed to identify perspective substituents hindering aggregation. The viscosity and density measurements further confirmed significant electrolyte's property tuning upon changing the structure of the lithium dicyanoimidazolidine, which is in line with the subsequent electrochemical measurements. Based on the complete gathered data, an extension via the 1,4-phenylene moiety along with the peripheral (O)CF₃-substitution proved to be useful strategy towards stabilized anions with a promising application in lithium-ion batteries.

Introduction

The growing demand for portable power sources has made lithium-ion batteries (LiBs) a cornerstone of modern technology, especially in consumer electronics. Nowadays, these batteries power everything from smartphones and laptops to electric vehicles, providing a lightweight and efficient energy solution.^{1–3} Despite the evolution in time and the recent advancements, the core technology remains unaltered, uses the lithium cation (Li⁺) as a primary charge carrier and profits from its reversible redox process taking place at the most negative potential (−3.04 V vs. SHE). Inorganic lithium salts such as LiPF₆ and LiBF₄ are considered conventional electrolytes, nevertheless suffering from some stability and safety issues.^{4–8} Considering the electrolyte a battery crucial part, these limitations drive the ongoing efforts to identify alternative salts. The vast majority of commercial LiBs utilize non-aqueous



electrolytes, dipolar aprotic solvents or ionic liquids are used in particular. The counterion to Li^+ acts as a stabilizing agent ensuring its dissociation, while being chemically inert and resistant to the battery's (electro)chemical environment. In overall, an idealized LIB's electrolyte combined with an organic solvent should: i) assure free movement of lithium (ions) between electrodes, thus ii) exhibit high ionic conductivity and minimal internal resistance, iii) possess electrochemical and iv) thermal stability (avoiding a degradation across the operating voltage/temperature window) and v) allow efficient formation of the solid electrolyte interphase (SEI) to ensure stable and efficient interaction between the electrolyte and the electrodes.^{9–11} Besides these fundamental criteria, the electrolyte should be further collector-friendly (non-corrosive), low viscose (within the operating temperature range), safe, environmentally friendly and cost-effective. Hence, various anions with the negative charge spread over the boron and nitrogen atoms and improved electrochemical performance and thermal robustness were developed. For instance, these include borates^{12–13} and imides such as lithium bis(fluorosulfonyl)imide (LiFSI)¹⁴ and bis(trifluoromethanesulfonyl)imide (LiTFSI).¹⁵ More recently, the concept of stable Hückel anions benefiting from a significant stabilization through covalently bonded electron-acceptor moieties has been extended towards nitrogen heterocycles.^{16–17} Five-membered 1*H*-imidazole and 1*H*-triazole proved to be the leading structural motives in designing such anions and lithium 2-trifluoromethyl-4,5-dicyanoimidazol-1-ide (**LiTDI**, Figure 1A) and lithium 4,5-dicyanotriazol-1-ide (**LiDCTA**, Figure 1A) represent the most typical examples. Whereas **LiDCTA** was utilized in polymer electrolytes,^{18–20} the synthesis of **LiTDI** was firstly invented by Bukowska et al.²¹ and its application in LIBs has further been extended by Niedzicki et al.^{21–25} and Berhaut et al.^{26–27} **LiTDI** has been reported to enable an efficient SEI formation,²⁸ while it may be also employed as a highly efficient additive to LiPF_6 -based electrolytes by suppressing its hydrolysis.²⁹ Triple-functionalization of the central imidazol-1-ide proved to be the key structural feature of **LiTDI**. Whereas the two electron-withdrawing cyano groups at the positions C4 and C5 improve the anion stability and weaken the Li^+ coordination, their linear arrangement allows to maintain a small molecular size with high solubility. The CF_3 group at position C2 enhances the intrinsic anion oxidation potential and lowers the HOMO energy, thus increases the HOMO–LUMO gap as compared to CH_3 - and H-substituted analogues.³⁰ Its electron-withdrawing property along with the bulky spatial arrangement, analogous to the isopropyl group, further enlarges the bond length between the ion pair, which reduces ion pair dissociation energy and results in higher ionic conductivity. The quantum-chemical computational studies showed that replacement of the CF_3 group by longer perfluorinated chains (e.g. C_2F_5 , C_3F_7) has only minimal impact.^{31–32}



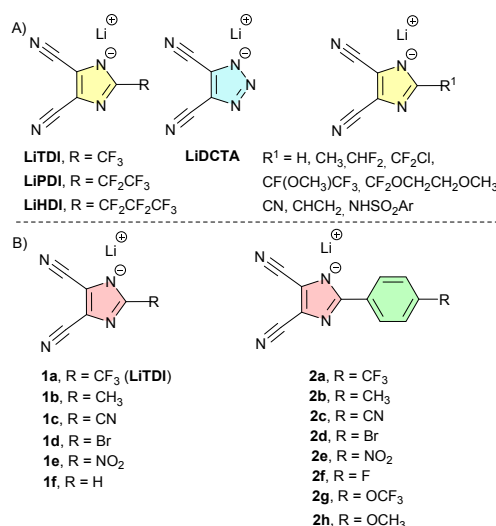


Figure 1. Molecular structure of known **LiTDI** (and its analogues **LiPDI** and **LiHDI**), **LiDCTA** and further C2-variation of imidazole-4,5-dicarbonitrile (A). Molecular structure of investigated (2-phenyl)-imidazole-4,5-dicarbonitrile derivatives **1** and **2** (B).

Compared to a practically unfeasible structural variation of **LiDCTA**, further modification of **LiTDI** has been carried out exclusively via an extension of the C2-appended trifluoromethyl group to perfluoroethyl (**LiPDI**) and perfluoropropyl ones (**LiHDI**), see Figure 1A.^{22–24,33} However, their practical application is constrained by the increased cost and availability of perfluorinated starting materials. When going from **LiTDI** to **LiHDI**, the elongated perfluoroalkyl chains bring an increase in viscosity due to the formation of stronger ion pairs and larger aggregation. However, the conductivity and ionicity of **LiHDI** and **LiPDI** in carbonates significantly exceed that of **LiTDI**. The structure of parent **LiTDI** has been further explored less extensively, especially by utilizing commercially available derivatives such as (2-amino)imidazole-4,5-dicarbonitriles. Screening the portfolio of currently available patents reveals that the original CF₃ group of **LiTDI** can be replaced by hydrogen, methyl,³⁴ additional perfluoroalkyl(ether),³⁵ cyano,³⁶ and vinyl groups³⁷ as well as sulfonamides bearing various Ar pendants (Figure 1A).³⁸ In addition, Rasmussen et al. reported tetracyanobimidazole however not intended for LIBs.³⁹

The chemistry of imidazole-4,5-dicarbonitrile (dicyanoimidazole, DCI) is well explored in our group as it has been introduced as a heterocyclic electron-withdrawing unit of various push-pull molecules with tuneable optoelectronic properties and manifold applications.^{40–44} Hence, we herewith demonstrate our systematic approach towards structural variation and property tuning of **LiTDI** by designing two series **1** and **2** (Figure 1B). Whereas the series **1a–f** consists of the imidazole-4,5-dicarbonitrile derivatives, the series **2a–h** is based on the parent 2-phenylimidazole-4,5-dicarbonitrile scaffold. An extension of the π -systems by 1,4-phenylene moiety^{45–46} is herewith proposed to stabilize the TDI[–] anion. Further stabilization is accomplished by appending substituents R featuring various inductive and/or mesomeric effects, including donors (CH₃, OCH₃, OCF₃, Br) and acceptors (CF₃, CN, NO₂). In series **2**, the substituents are appended in position 4 of the phenyl ring to utilize the whole conjugated pathway and maintain the molecular symmetry. 1,4-Phenylene moiety in **2a–h** also assures better distancing of the imidazolide anion and the R functional group and potentially reduces the crystal lattice energy of the lithium salt, especially when R contains electronegative

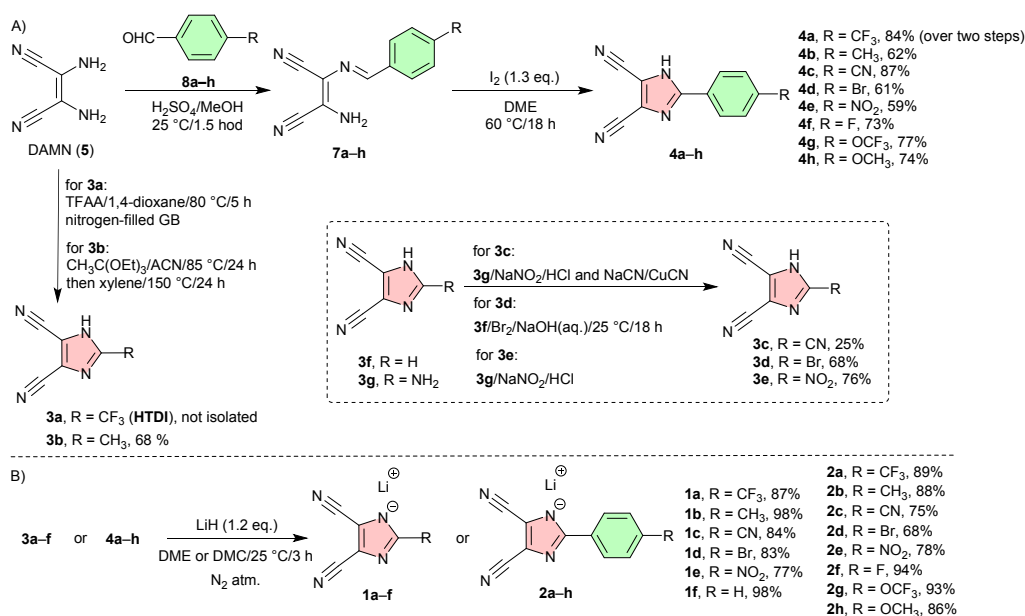


fluorine.¹⁰ Moreover, fluorine-containing groups generally improve solubility and stability in solution.

Results and discussion

Synthesis

The lithium salts **1a–f** and **2a–h** were synthesized via a general two-step protocol as outlined in Scheme 1 (see the Supplementary information (SI) for the complete synthetic details). Whereas the parent 1*H*-imidazoles **3f** and **3g** are commercially available, remaining 1*H*-imidazole-4,5-dicarbonitrile derivatives **3** and **4** were synthesized. Cyclocondensation of diaminomaleonitrile (DAMN, **5**) with trifluoroacetic anhydride (TFAA) or triethyl orthoacetate afforded 1*H*-imidazoles **3a** and **3b**, while electrophilic bromination of **3f** or diazotization/Sandmeyer reactions with **3g** provided derivatives **3c**, **3d** and **3e**. DAMN can be also treated with substituted benzaldehydes **8a–h** towards imine intermediates **7a–h** that were cyclized under oxidative condition to 1*H*-imidazoles **4a–h**. This way prepared *NH*-acids **3** and **4** were subsequently reacted with LiH to yield the desired lithium salts **1** and **2** (Scheme 1B). The lithiation and all further treatment of **1** and **2** were carried out in a nitrogen-filled glovebox due to their high hygroscopicity.



Scheme 1. Synthetic strategy towards 1*H*-imidazoles **3** and **4** (A) and their neutralization to lithium salts **1** and **2** (B).

Substituent effects

To quantify the varied substitution pattern in 1*H*-imidazoles **3** and **4** and extent of the charge delocalization in the corresponding imidazolides **1** and **2**, dissociation constants (pK_a) were determined via potentiometric titration (Table 1).^{47–49} Two solvents were used, protic methanol (MeOH) and aprotic acetonitrile (ACN). Having the same 1*H*-imidazole-4,5-dicarbonitrile moiety in all derivatives **3**, the variation in the acidity is ascribed to the electronic effects of the substituent appended at the position C2. When going from the parent HTDI (**3a**, pK_a = 3.71/10.96 in MeOH/ACN) to C2-unsubstituted (**3f**) and methyl-substituted (**3b**) derivatives, the acidity decreases by five/seven order of magnitude as a result of removing the CF₃ inductive acceptor and attaching hydrogen or the CH₃ donor.



Table 1. The average dissociation constants ($\overline{pK}$) and their standard deviations (s) of 1*H*-imidazoles **3** and **4**.

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1 <i>H</i> -imidazole	$\overline{pK} (\pm s)^a$	
	MeOH	ACN
3a	3.71 (0.09)	10.96 (0.11)
3b	8.63 (0.05)	17.92 (0.08)
3c	3.46 (0.03)	9.70 (0.02)
3d	4.94 (0.08)	12.68 (0.13)
3e	9.45 (0.09)	21.09 (0.17)
3f	7.73 (0.04)	16.60 (0.06)
4a	6.78 (0.06)	15.19 (0.12)
4b	7.87 (0.04)	16.98 (0.10)
4c	6.43 (0.07)	14.65 (0.11)
4d	7.48 (0.02)	16.56 (0.04)
4e	6.44 (0.07)	14.69 (0.09)
4f	7.56 (0.02)	16.73 (0.08)
4g	7.52 (0.03)	16.52 (0.09)
4h	8.01 (0.06)	17.18 (0.09)

^a Determined experimentally via potentiometric titration using Bu₄N⁺HO[−] base and benzoic acid as standard ($pK_0 = 9.41$ (MeOH)⁵⁰ and 20.70 (ACN)⁵¹. An average value of 3–5 experiments.

The highest acidity has been measured for tricyanoimidazole **3c**, which reflects attachment of the linear mesomeric C≡N acceptor. Surprisingly, attaching strongly withdrawing nitro group (**3e**) increases pK_a to 9.45/21.09. Based on the DFT-optimized structure of **3e** and **1e** (Figure S3), the reduced acidity is likely ascribed to a strong intramolecular hydrogen bonding between the imidazole N1 and the oxygen atom of the nitro group. In general, the transmission effect of the additional 1,4-phenylene moiety strongly reduces differences in the acidity and decreases pK_a by three/four order of magnitude (e.g. 6.78/15.19 and 3.71/10.96 for **4a** and **3a**). Hence, the acceptor-substituted 1*H*-imidazoles became less acidic (e.g. **4a** and **4c**) but the acidity of the donor-substituted ones is pronounced (e.g. **4b**). The pronounced acidity is also encountered for 4-nitrophenyl derivative **4e** confirming the aforementioned assumption on hydrogen bonding of the NO₂ group in **3e**, which is not possible in the extended **4e**. The substituent effects in the series **4a–h** were quantitatively evaluated using Hammett equation.⁵² Well-fitting linear regressions of pK_a vs. σ_p (the reaction constant $\rho = -1.55/-2.552$, the correlation coefficient $r = 0.968/0.955$ and the number of points $N = 31/30$) were obtained in both solvents (MeOH/ACN). The substituent effects in 2-phenylimidazolides **2a–h** were also investigated using ¹³C chemical shifts of the imidazolidine C2 carbon atom (Tables S1). The resulting good linear regressions of $\delta(^{13}\text{C-C2})$ vs. σ_p ($r = 0.981$, $N = 7$, **2b** excluded as an outlier) as well as pK_a of **4a–h** vs. $\delta(^{13}\text{C-C2})$ of **2a–h** ($r = 0.911/0.859$, $N = 34/35$ in MeOH/ACN) confirm an analogous transfer of the electronic effects of the appended substituents R to the imidazol(id)e-4,5-dicarbonitrile moieties, while the electron donating/withdrawing behaviour of the R substituents resembles those of 4-substituted benzoic acid derivatives.



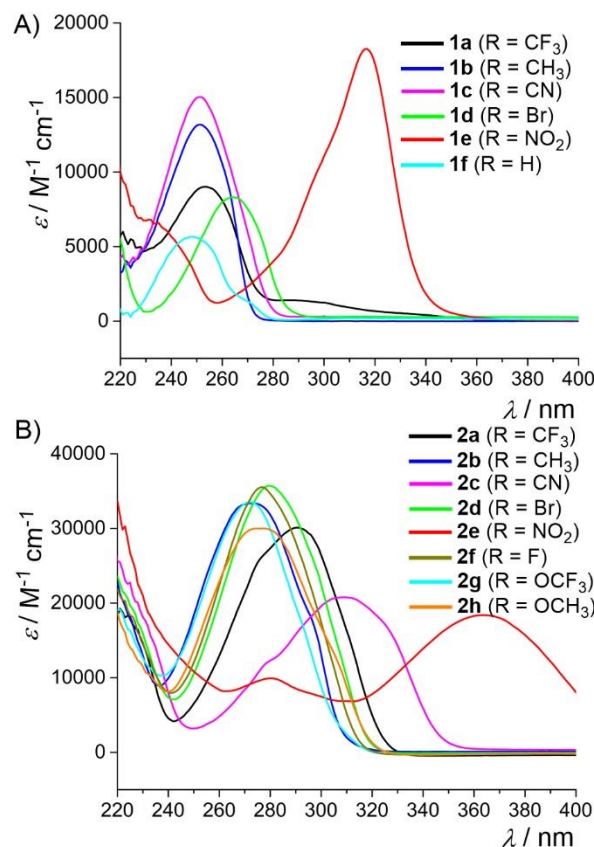


Figure 2. Electronic absorption spectra of imidazolidines **1** (A) and **2** (B) in DME/DMC (1:1), $c = 5$ and 2×10^{-5} M)

The electronic absorption spectra of lithium salts **1** and **2** (Figure 2; see also Table 2 for the longest wavelength absorption maxima $\lambda_{\max}^A$ and the molar absorption coefficients ε) further supports the potentiometric measurements. Whereas in **1a–c** and **1f** are the longest absorption bands positioned nearly identically and differ mostly in the molar absorption coefficient as a result of attaching either electron acceptor or donor groups (CF_3 , CH_3 , CN or H), nitroderivative **1e** showed exceptionally red-shifted absorption maxima, which points to its different structural arrangement. The substituent effects are even more pronounced in the series **2**, where an insertion of the 1,4-phenylene moiety generally red-shifts the absorption maxima by 20–60 nm. The position of the longest wavelength absorption band depends further on the substitution, the electron acceptors such as CF_3 (**2a**), CN (**2c**) and NO_2 (**2e**) induce the most bathochromically shifted ones. In summary, employing the 1,4-phenylene moiety along with the varied (C2/*para*-)substitution (**a–h**) seems to be a useful strategy to fine-tune the acidity of 1*H*-imidazoles **3** and **4**. The most stabilized conjugated bases (imidazolidines **1/2**) can be generated from 1*H*-imidazoles **3c/3a** and **4a/4c/4f** bearing $\text{CF}_3/\text{CN}/\text{NO}_2$ acceptors.

Preparation of electrolytes

Carbonates, (poly)ethers (glymes), esters, amides, sulfones and ionic liquids belong to the most widely employed organic solvents in LIBs.^{3,53–55} As an essential prerequisite to their successful application, solubility of lithium salts **1** and **2** in dimethyl carbonate (DMC), 1,2-dimethoxyethane (DME) and commercial SolMIX were firstly screened (Table 2). In all solvents at 25 °C, the benchmark LiTfD and LiPF₆ showed the largest solubility and up to 1.5/4.8/3.3 and 7.9/1.3/4.9M solutions can be prepared. The data gathered in DMC further



indicates that the salts **1b–f** and **2a–h** are much less soluble or even insoluble with the exception for **2a** and **2g** bearing CF₃ and OCF₃ substituents (solubility up to ca 0.3 M). In contrast to DMC, the solubility in pure DME is significantly enhanced and ranges from 0.4 to 2.8 M, including some derivatives practically insoluble in DMC. When comparing salts **1** and **2**, the additional 1,4-phenylene linker in **2** brings enhanced solubility, which further varies based on the substitution. Finally, 1:1 volumetric mixture of DME and DMC and 0.1, 0.3, 0.6 and 1M concentrations were chosen for the further investigation. The electrolytes were freshly prepared in a nitrogen-filled glovebox by dissolving the particular lithium salt in DME and subsequently diluting the solution with DMC. The obtained solubility profiles also served as a primary filter in identifying derivatives with a potential for a practical application (**1a**, **2a/b** and **2f–h**), while compounds exhibiting poor solubility were excluded from the subsequent investigations. The moisture content is crucial for the overall performance of the prepared electrolytes and, therefore, we have determined the water content via a coulometric titration (Karl Fischer titrator) to be under 30 ppm. Compared to benchmark LiPF₆, imidazolides **1a**, **2a** and **2g** showed enhanced thermal robustness as measured by DSC (Table 2 and the ESI).

Table 2. Summary properties of the studied imidazolides **1** and **2**.

Der.	Solubility [g·L ⁻¹]/[mol·L ⁻¹] ^a			$\lambda_{\text{max}}^{\text{A}}$ [nm/eV] ^b	ϵ [M ⁻¹ cm ⁻¹] ^b	T_d [°C] ^c	η [mPa·s] ^d	ρ [g·cm ⁻³] ^e	E_a [kJ·mol ⁻¹] ^f	r_s [nm] ^g	r_{eff} [nm] ^h	D [×10 ⁻¹⁰ m ² ·s ⁻¹] ⁱ	κ [mS·cm ⁻¹] ^j
	DMC	DME	SolMIX										
1a	300/1.56	924/4.81	640/3.30	253/4.90	9000	240	0.884	1.017	14.10	0.548	0.576	4.222	6.33
1b	8/0.06	insoluble	-	251/4.91	13200	-	-	-	-	-	-	-	-
1c	insoluble	300/2.01	-	251/4.91	15000	-	-	-	-	-	-	-	-
1d	10/0.05	260/1.30	-	264/4.70	8300	-	-	-	-	-	-	-	-
1e	6/0.04	230/1.36	-	317/3.91	18300	-	-	-	-	-	-	-	-
1f	6/0.05	insoluble	-	248/5.00	5700	-	-	-	-	-	-	-	-
2a	100/0.37	380/1.42	330/1.23	291/4.26	30100	260	1.037	1.021	14.21	0.557	0.646	3.205	6.20
2b	insoluble	600/2.80	20/0.09	273/4.54	33400	-	1.096	1.008	14.47	0.528	0.670	2.925	3.77
2c	insoluble	100/0.44	insoluble	310/4.00	20800	-	-	-	-	-	-	-	-
2d	insoluble	720/2.58	80/0.29	280/4.43	35700	-	-	-	-	-	-	-	-
2e	5/0.02	340/1.39	190/0.78	363/3.42	18400	-	-	-	-	-	-	-	-
2f	insoluble	440/2.02	40/0.18	272/4.56	33500	-	1.060	1.021	14.07	0.568	0.655	3.092	5.93
2g	90/0.32	800/2.82	420/1.48	277/4.48	35500	190	1.047	1.025	14.34	0.533	0.650	3.154	6.23
2h	insoluble	440/1.91	insoluble	277/4.48	30000	-	1.024	1.007	14.85	0.617	0.641	3.274	3.04
LiPF ₆	1200/7.90	200/1.32	740/4.87	-	-	180	1.211	1.024	-	-	0.711	2.494	7.87

^a Determined in dimethyl carbonate (DMC), 1,2-dimethoxyethane (DME) and SolMIX (a commercial mixture of carbonates DMC:EC:EMC, (1:1, v/v) at 25 °C using a temperature-tempered ultrasonic bath. ^b The longest wavelength absorption maxima ($\lambda_{\text{max}}^{\text{A}}$)/the molar absorption coefficient (ϵ) measured in DME/DMC (1:1) at the concentrations 5×10⁻⁵ (**1**) and 2×10⁻⁵ (**2**). ^c Temperature of endothermic/exothermic decomposition under inert atmosphere determined by DSC (see Table S2 and Figure S4). ^d The experimentally measured dynamic viscosity (0.6 M in DME/DMC (1:1)) at 20 °C. ^e The experimentally measured densities (0.6 M in DME/DMC (1:1)) at 20 °C. ^f The activation energy of viscous flow determined from the Arrhenius equation. ^g The effective solute radius at 20 °C. ^h The effective hydrodynamic radius at 20 °C ($c = 0.6$ M). ⁱ Self-diffusion coefficient at 20 ° and $c = 0.6$ M. ^j The experimentally measured specific conductance (0.6 M in DME/DMC (1:1)) at 20 °C. The electrochemical stability windows of the investigated electrolytes were evaluated using linear sweep voltammetry (LSV) from 0 to 5 V, see Figure 6 below.

Associations and aggregations of electrolytes

Having addressed the formulation of electrolytes, we focused on behaviour of the selected lithium salts in the DME/DMC solution. We address a molecular association and a formation of lower-order aggregates, such as dimers and trimers, as a function of concentration (Figure 3 and Figures S5–12 in the ESI). The concentration of LiTDI (**1a**) was gradually increased from 10⁻⁵ M and the corresponding electronic absorption spectra were measured (Figure 3A). We focused especially on tailing of the low-energy region, which is indicative of Mie scattering effect on the formed aggregates.^{56–58} The tailing is evident already at 5×10⁻⁴ M, which is significantly lower concentration than reported for the Raman/FTIR analysis (0.1–0.63 M).³³ At 5×10⁻² M, a structured new band appearing at around 310–330 nm is clearly visible pointing to a formation of various aggregate types. In contrast to this observation, extended salt **2a**

(Figure 3B) showed only gradually increasing absorbance without any concomitant tailing. Thus, up to 5×10^{-2} M concentration, no formation of aggregates was observed. While **2b**, **2d** and **2h** (Figures S7–8 and S11) behave similarly, 4-fluoro (**2f**, Figure 3C) and 4-trifluoromethoxy (**2g**, Figure S10) derivatives began to form aggregates at around 10^{-3} and 10^{-2} M; the one/two order of magnitude higher concentration than benchmark **LiTDI**.

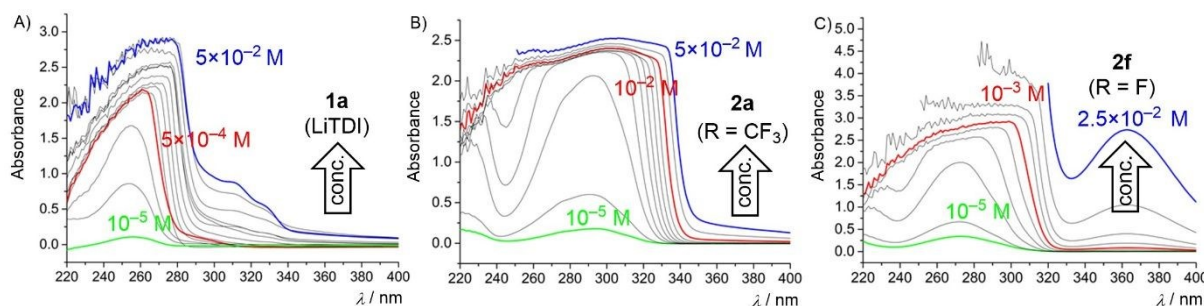


Figure 3. Representative electronic absorption spectra of lithium salts **1a** (A), **2a** (B) and **2f** (C) with gradually increased concentration (DME:DMC 1:1).

Viscosity and density of electrolytes

Dynamic viscosity and density were measured over the temperature range of -10 to 60 °C with the constant solvent system (DME/DMC 1:1). Table 2 lists the recorded values measured for 0.6 M solutions of **1a**, **2a–b** and **2f–2h** at 20 °C, while Figure 4 shows exponentially decreasing viscosity and density with the increasing temperature. A rather linear trend is observed especially for the density of the electrolytes of lower concentration (Figure S12). To assess temperature dependence of the dynamic viscosity, the activation energy of viscous flow (E_η) was calculated (Table 2 and S3) by linearly fitting η to $1/T$ with the aid of the Arrhenius equation (see the ESI and Figure S13).⁵⁹ The raising E_η values of all derivatives with the increased concentration reflect pronounced intermolecular interactions hindering an efficient viscous flow and further corroborates the aforementioned formation of the clusters and aggregates. However, the viscous flow of the benchmark **LiTDI** substantially depends on the concentration ($\Delta E_\eta = 2.77$ kJ·mol⁻¹; Table S3), which contrasts the relatively steady E_η values measured for **2a** ($\Delta E_\eta = 0.97$ kJ·mol⁻¹). As compared to **LiTDI**, the most efficient viscous flow was recorded for extended derivatives **2a**, **2b**, **2g** and **2h** (0.3M) possessing a diminished tendency to aggregate as determined by the UV/Vis spectroscopy (see above). Using Eyring model,⁶⁰ the measured dynamic viscosity has further been recalculated to activation parameters of viscous flow such as Gibbs energy of activation ($\Delta G^\ddagger$), enthalpy of activation ($\Delta H^\ddagger$) and entropy of activation ($\Delta S^\ddagger$), see Tables S4–5 and Figure S14. The gathered data point to an increased viscous flow with the raised temperature, while the increasing concentration has expectedly the opposite effect. As compared to other derivatives, salt **2a** showed significantly reduced $\Delta H^\ddagger$ and $\Delta S^\ddagger$ values at 0.3M, which may indicate weaker intermolecular interactions and a diminished degree of order, while the positive entropy also points to a disorganization during the viscous flow (a disruption of the molecular interactions).



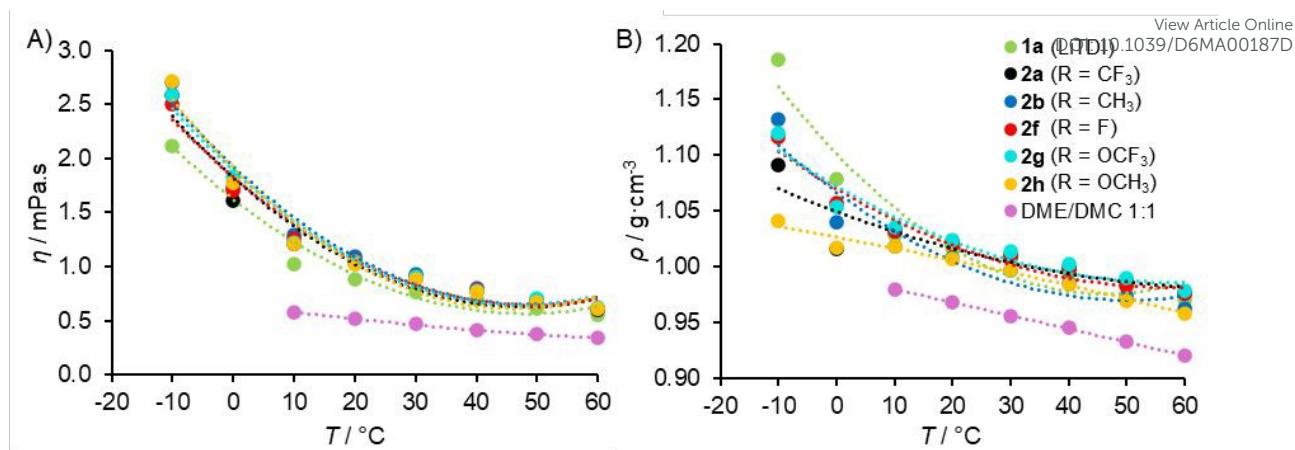


Figure 4. The dynamic viscosity η (A) and density ρ (B) of imidazolides **1a**, **2a–h** and **2f–h** as a function of the temperature (0.6 M solutions in DME/DMC 1:1).

The ionic interactions and solvation effects of the investigated electrolytes were also examined using Jones-Dole-Kaminsky (JDK) equation^{59,61–62} that allows to determine the solute effective radius r_s (Tables 2 and S6, Figure S15). Whereas the calculations at 20 °C revealed rather similar r_s values for all derivatives (Table 2), increasing temperature reduces the radii due to a weakened ion-solvent interaction (a thermal disruption of the solvation shell), see Table S7. A similar trend is seen for the B parameter, which also relates to the molar volume. When comparing the benchmark **LiTf** and **2a**, nearly identical B parameters were calculated within the temperature range of 10–60 °C. The electrolyte **2a** shows relatively steady D parameter, compared to its gradually increasing value for **LiTf**, which points to a nearly temperature independent ion-ion interaction and highlights the effect of the 1,4-phenylene moiety. Considering the whole series of electrolytes, **2b** (R = CH₃) showed the lowest B and the highest D parameters, suggesting limited solvation and enhanced ion pairing, while the situation is completely opposite for **2h** (R = OCH₃). This obvious difference demonstrates influence of the coordinating ether moiety (-O-) in affecting the ion-solvent and ion-ion interactions. Employing complementary Einstein's theory of dilute suspensions,^{63–64} the effective hydrodynamic radius r_{eff} was also evaluated (Table 2 and Figure S16). In contrast to the JDK model focused on localized ion-solvent and ion-ion interactions, the Einstein model includes solvation shell and aggregates, which is reflected in generally larger r_{eff} values as compared to r_s ones. However, both quantities follow the same trend within the series **2**. The effective hydrodynamic radius has been used to calculate self-diffusion coefficient D (Table 2 and Table S8), characterizing the intrinsic mobility of solvated ions under thermal agitation. Unlike the pulsed-field gradient NMR measurements,^{33,59} providing separate diffusion coefficients for cations and anions, this indirect Stokes-Einstein method⁶⁵ yields a global average that assumes spherical symmetry and ideal behaviour but is generally considered useful for a comparative study. A comparison of the NMR-estimated $D = 4.25 \times 10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$ of **LiTf** (EC:EMC 3:7, 0.6 M, 40 °C)³³ and our $D = 4.22 \times 10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$ of **LiTf** (DME:DMC 1:1, 0.6 M, 20 °C) implies a very good agreement of both methods. In general, the obtained self-diffusion coefficients (Table S8) are highest for **1a**, which reflects its small size compared to derivatives **2** extended by 1,4-phenylene moiety. However, in diluted 0.1 M solutions, the D values for **1a** and **2a** are very close (9.763 vs. $9.288 \times 10^{-10} \text{ m}^2 \cdot \text{s}^{-1}$). The mobility of ions linearly increases with the raising temperature (Figure S17), while the increasing viscosity/concentration has the opposite effect (Figure S18).



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As expected, the conductometric behaviour of **LiTDI** shows a systematic increase in the ionic conductivity as a function of the raised concentration (0.1 to 1.0 M, Table S9 and Figure S19). However, the curvature above $c = 0.6$ M suggests a diminishing rate caused probably by the growing viscosity and the aggregation phenomena (see above), both reducing the ion mobility. The specific conductance of **1a**, **2a–b** and **2f–2h** measured uniformly for 0.6M solutions (Table 2) reveals **2a** and **2g** ($\kappa = 6.20$ and $6.23 \text{ mS}\cdot\text{cm}^{-1}$) with nearly identical values as measured for the benchmark **LiTDI** ($\kappa = 6.33 \text{ mS}\cdot\text{cm}^{-1}$). Thus, the extension of the π -system has not negatively influenced the conductivity. On the contrary, removing the fluorine atom(s) from the R-substituent has detrimental effect, especially for **2b** and **2h** ($\kappa = 3.77$ and $3.04 \text{ mS}\cdot\text{cm}^{-1}$).

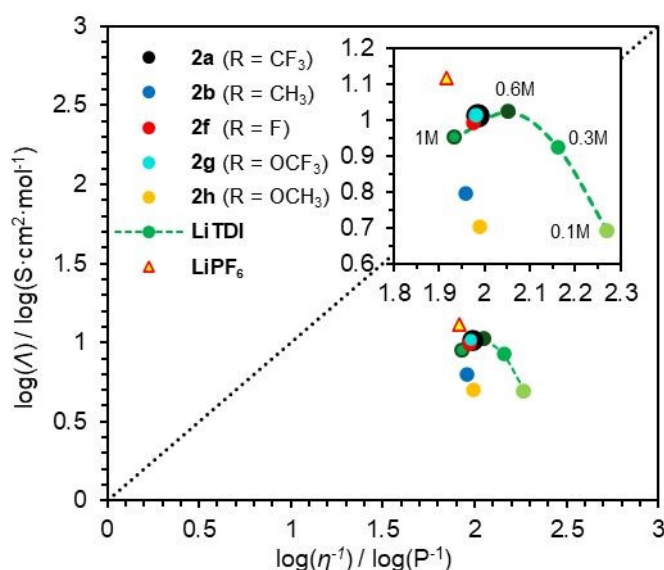


Figure 5. Molar conductivity (Λ) versus inverse dynamic viscosity (η^{-1}) for lithium imidazolidine-based electrolytes (**2a,b** and **2f–2h** at 0.6 M; **1a** at 0.1, 0.3, 0.6 and 1.0 M) compared to LiPF_6 in DME/DMC (1:1) at 20 °C. The dotted line represents the ideal Walden line (fully dissociated KCl). An extended area is also provided as an inset.

To assess the ionic dissociation behaviour of lithium imidazolidine salts, Walden plot has been further constructed (Figure 5 and Table S10). In general, all tested electrolytes deviate below the ideal KCl line, indicating an incomplete dissociation and the presence of ion pairing or aggregation. The data for **LiTDI** at low concentrations (0.1–0.3 M) substantially deviates from the ideal Walden function, whereas the 0.6 and 1 M solutions lie closer. This trend is accompanied by an atypical increase in the ionicity with the raised concentration (from 2.6 to 9.3 %) and a corresponding reduction in the vertical offset ΔW from -1.580 to -1.030 . This observation contrasts the behaviour generally reported for carbonate-based electrolytes^{26,33} and suggests that, in an ether–carbonate medium, a gradual salt addition enhances ionic mobility or partial dissociation. According to the study of Forsyth et al.,⁶⁶ using a binary ether-rich mixture leads to a breakdown of the extended ion networks and a faster ion-exchange, most probably due to σ -donor character of the oxygen atoms of DME.⁶⁷ In addition, we would further highlight the lower viscosity/density of DME ($\eta/\rho = 0.4341/0.8665 \text{ mPa}\cdot\text{s}^{-1}/\text{g}\cdot\text{cm}^{-3}$) as compared to DMC ($\eta/\rho = 0.585/1.0635 \text{ mPa}\cdot\text{s}^{-1}/\text{g}\cdot\text{cm}^{-3}$), while their 1:1 mixture possesses $\eta/\rho = 0.5133/0.9679 \text{ mPa}\cdot\text{s}^{-1}/\text{g}\cdot\text{cm}^{-3}$ at 20 °C. Anyway, a complete dissociation has not been achieved independently on the concentration, which is in line with the current literature data on similar Hückel-type lithium salts.^{16,68} When focusing on the structural variation in **2a–b** and



2f–h at the steady concentration (0.6 M), **2a** and **2g** showed the smallest deviations from the ideality (−0.970 and −0.964) corresponding to the highest ionicity (over 10.7%). On the contrary, nonfluorinated salts **2b** and **2h** exhibited pronounced deviations (−1.162 and −1.285) and lower ionicity (6.9 and 5.2%). This suggest higher concentration of free extended anions **2a**, **2f** and **2g** as compared to the benchmark **1a** as well as the nonfluorinated imidazolides **2b** and **2h**.

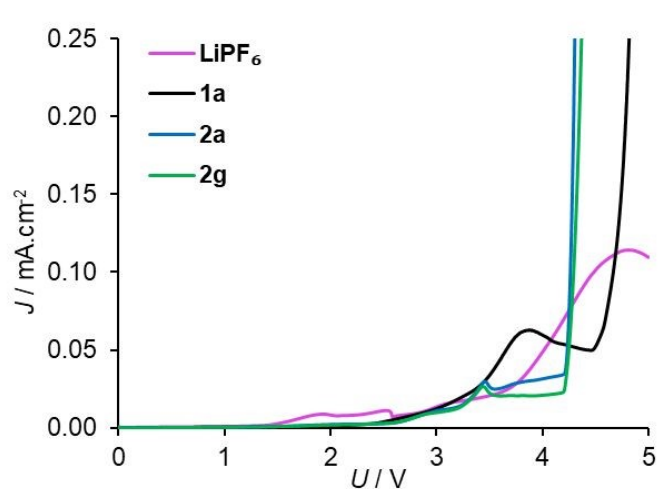
Table 3. Parameter estimated for Randles equivalent circuit model from the EIS data measured for **1a**, **2a** and **2g** in DME:DMC (1:1).

Salt (concentration [mol·l ⁻¹])	R_1 [Ohm]	R_2 [Ohm]	W_1 [Ω·s ^{-1/2}]	C_1 [F]
1a (0.1)	1553	422.3	2330	3.888×10^{-9}
1a (0.3)	378.0	0.023	1492	2.459×10^{-5}
1a (0.6)	150.9	0.007	1575	3.391×10^{-5}
1a (1.0)	106.8	0.005	1575	3.737×10^{-5}
2a (0.6)	242.3	0.016	1561	3.386×10^{-5}
2g (0.6)	240.2	0.013	1253	2.952×10^{-5}

Alongside the conductometry, the EIS was performed within the range of 50 – 200000 Hz using conductivity probe. The electrochemical impedance analysis revealed a pronounced dependence of the Randles circuit parameters of **LiTDI** within the concentration range of 0.1–1.0 M (Table 3). Increasing the concentration systematically reduces the solution resistance (R_1) from 1553 to 106.8 Ω, reflecting an enhanced ionic conductivity. The charge-transfer resistance (R_2) decreases by more than five orders of magnitude (422→0.005 Ω), indicating a substantial improvement in the interfacial reaction kinetics at higher ionic strength. The double-layer capacitance (C_1) rises from a few nF to several tens of μF, which is consistent with a compression of the electrical double layer and shortening of the Debye length. The Warburg coefficient (W_1) is highest at 0.1 M and stabilizes thereafter, suggesting the diffusion limitations are significant only at low concentration of **LiTDI**. These observations confirm that the transport and interfacial processes dominate at dilute conditions, whereas the system approaches an ideal behaviour with minimal polarization losses at higher concentrations. The EIS measurements of **2a** and **2g** (0.6 M) point to a slightly lower bulk conductivity ($R_1 \sim 240$ Ω) compared to **LiTDI** ($R_1 = 150.9$ Ω at the same concentration) but their charge-transfer resistances remain very low (0.016/0.013 Ω for **2a/2g**) and comparable to **LiTDI**, suggesting that the interfacial kinetics are not significantly compromised upon extending the π-system or embedding the oxygen atom (CF₃→OCF₃). The nearly identical double-layer capacitance of both salts and **LiTDI** (~ 30–34 μF) confirms similar surface characteristics and effective electrode wetting. The lowest Warburg coefficient deduced for **2g** (1253 Ω·s^{-1/2}), as compared to **2a** and **LiTDI** (1561 and 1575 Ω·s^{-1/2}), may indicate slightly improved ion diffusion for **2g**.

Linear sweep voltammetry (LSV) was employed to evaluate the electrochemical stability window of imidazolides **2a**, **2g** and **LiTDI** and compare them with the benchmark LiPF₆ in DME:DMC solvent mixture (Figure 6). All salts exhibited very low background current densities up to approximately 4 V, indicating negligible parasitic reactions and good electrochemical stability within this potential range. When increasing the potential above 4 V, both **2a** and **2g** show a sharp current increase at around 4.2 V, suggesting a slightly lower oxidative limit compared to **LiTDI** (~4.6 V). LiPF₆ displays a similar trend to **LiTDI** but with a slightly earlier and more moderate increase in the current density.





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Figure 6. The LSV for **LiTDI**, **2a** and **2g** along with the benchmark LiPF_6 [0.6 M solutions in DME:DMC (1:1)].

Conclusion

In this work, we synthesized fourteen lithium (2-phenyl)-4,5-dicyanoimidazolides with varied peripheral pendants (CF_3 , CH_3 , CN , Br , NO_2 , H , F , OCH_3 and OCF_3). Starting from the diaminomaleonitrile, the used synthetic method relies on a straightforward cyclocondensation/oxidation followed by the reaction with lithium hydride. The substituent effects in the prepared 1*H*-imidazoles have been examined by potentiometrically-determined dissociation constants proving large tunability of their acidity. The pK_a values measured in methanol range from 3 to 10, while even large differences were found in acetonitrile. The substituent effects of the imidazolides **1** and **2** investigated by ^{13}C NMR and electronic absorption spectra further confirmed stability of the corresponding conjugated bases as a function of their substitution. The CF_3 , CN and NO_2 groups proved to be the most beneficial in this respect. Traditional dimethyl carbonate along with 1,2-dimethoxyethane solvents were utilized to prepare electrolytes, the latter solvent showed significantly enhanced solubilization of the prepared salts **1** and **2**. The particular derivatives **1a**, **2a/b** and **2f–h** were identified as sufficiently soluble in DMC:DME 1:1 mixture and their solutions were further examined for the association and aggregation phenomena. By employing electronic absorption spectra, substantially more sensitive methodology addressing aggregation is presented. Especially the extended 2-(4-trifluoromethylphenyl)-4,5-dicyanoimidazolidine **2a** showed no aggregation up to 5×10^{-2} M concentration, which is in line with its nearly concentration/temperature-independent viscous flow and ion-ion interactions. Despite their comparable molecular volume, effective hydrodynamic radius and self-diffusion coefficient, the benchmark **LiTDI** aggregates already at 5×10^{-4} M and possesses a concentration/temperature-dependent viscous flow. Compared to **LiTDI** ($\kappa = 6.33 \text{ mS} \cdot \text{cm}^{-1}$), the extended derivatives **2a** and **2g** bearing CF_3 and OCF_3 peripheral substituents possess nearly the same specific conductance of 6.20 and 6.23 $\text{mS} \cdot \text{cm}^{-1}$. A significant drop in the κ values is seen upon removing the fluorine atom(s). The Walden plot revealed the highest ionicity ($\sim 10.7\%$) for **2a** and **2g**, larger than for **LiTDI** (9.3%), while the EIS and LSV measurements suggest slightly higher ohmic resistance (~ 240 vs. 151 Ω) and slightly lower oxidative limit (4.2 vs. 4.6 V) as compared to **LiTDI**. However, the charge-transfer resistance and the double-layer capacitance are nearly identical, while the lowest Warburg coefficient of **2g** implies again higher ion diffusion. In summary, the systematically evaluated structure of lithium imidazolides **1** and **2** allowed to identify the π -



system extension (1,4-phenylene moiety) and variation of the peripheral substituents (R) as a suitable tool to design stabilized anions and LIB's electrolytes. Whereas the π -system extension and substitution by electron withdrawing R-substituents help to delocalize the imidazolidine electron excess, the fluorine-bearing substituents generally improves solubility of the ionic salts in an aprotic media of low polarity. The new imidazolides **2a** and **2g** maintain good interfacial performance and diffusion behaviour and their wide electrochemical window makes them suitable for LIBs based on LFP/LMFP or NMC and graphite with the maximal potential of 4.2 V. Moreover, the synthesis of **2a/g** starts from the inexpensive and readily available 4-substituted benzaldehydes **8a/g** affording the imine intermediates **7a/g** smoothly, while the subsequent oxidation to **4a/g** utilizes inexpensive molecular iodine. This contrasts to the synthesis of **LiTDI** using large excess of toxic TFAA, elevated temperature and subsequent laborious isolation.

Author contributions

Daniel Pokorný: Investigation, Methodology, Formal analysis, Writing – original draft. **Tomáš Syrový:** Investigation, Methodology, Funding acquisition, Writing – original draft. **Milan Klikar:** Investigation. **Patrik Pařík:** Investigation. **Zuzana Burešová:** Investigation. **Lenka Řeháčková:** Investigation. **Filip Bureš:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Conflict of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the Supplementary Information. Supplementary information: Synthesis, Dissociation constant and substitution effects, thermal characterization, aggregation, viscosity and density, electrochemical measurements, NMR and HR-MS spectra. See DOI: [URL – format <https://doi.org/DOI>]

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Data availability

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The data supporting this article have been included as part of the Supplementary Information. Supplementary information: Synthesis, Dissociation constant and substitution effects, thermal characterization, aggregation, viscosity and density, electrochemical measurements, NMR and HR-MS spectra. See DOI: [URL – format <https://doi.org/DOI>]

